Journal of Molecular Spectroscopy 322 (2016) 22–28

Contents lists available at ScienceDirect

Journal of Molecular Spectroscopy

journal homepage: www.elsevier.com/locate/jms

Accurate dipole moment curve and non-adiabatic effects on the high resolution spectroscopic properties of the LiH molecule

Leonardo G. Diniz [a,*], Nikita Kirnosov [b], Alexander Alijah [c], José R. Mohallem [d], Ludwik Adamowicz [e]

[a] Coordenação de Ciências, Centro Federal de Educação Tecnológica de Minas Gerais, 30.421-169 Belo Horizonte, MG, Brazil
[b] Department of Physics, The University of Arizona, Tucson, AZ 85721, USA
[c] Groupe de Spectrométrie Moléculaire et Atmosphérique, GSMA, UMR CNRS 7331, University of Reims Champagne-Ardenne, U.F.R. Sciences Exactes et Naturelles, 51687 Reims Cedex 2, France
[d] Laboratório de Átomos e Moléculas Especiais, Departamento de Física, ICEx, Universidade Federal de Minas Gerais, PO Box 702, 31270-901 Belo Horizonte, MG, Brazil
[e] Department of Chemistry and Biochemistry, The University of Arizona, Tucson, AZ 85721, USA

ARTICLE INFO

Article history:
Received 21 January 2016
In revised form 29 February 2016
Accepted 1 March 2016
Available online 2 March 2016

Keywords:
Dipole moment
Einstein coefficients
Non-adiabatic effects

ABSTRACT

A very accurate dipole moment curve (DMC) for the ground $X^1\Sigma^+$ electronic state of the ^{7}LiH molecule is reported. It is calculated with the use of all-particle explicitly correlated Gaussian functions with shifted centers. The DMC – the most accurate to our knowledge – and the corresponding highly accurate potential energy curve are used to calculate the transition energies, the transition dipole moments, and the Einstein coefficients for the rovibrational transitions with $\Delta J = -1$ and $\Delta v \leqslant 5$. The importance of the non-adiabatic effects in determining these properties is evaluated using the model of a vibrational R-dependent effective reduced mass in the rovibrational calculations introduced earlier (Diniz et al., 2015). The results of the present calculations are used to assess the quality of the two complete linelists of ^{7}LiH available in the literature.

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1. Introduction

The LiH molecule has been the focus of several theoretical and experimental works in the last decades (see, for example, Stwalley and Zemke [1]). This smallest neutral heteronuclear molecule with just four electrons has been a model system for testing state-of-the-art *ab initio* electronic structure calculations (see, for example, Tung et al. [2] and Holka et al. [3]). Besides, its small mass and the existence of four stable isotopologues, ^{7}LiH, ^{6}LiH, ^{7}LiD, and ^{6}LiD, makes it also a good model to study non-adiabatic (or non-Born–Oppenheimer) effects. An avoided crossing between the low-lying potential energy curves (PECs) corresponding to the ground $X^1\Sigma^+$ state and the first excited $A^1\Sigma^+$ state around $5.7\,a_0$ makes this study particularly interesting.

LiH is an important system in astrophysics, as it plays a fundamental role in the cooling process of primordial gas clouds in the early universe [4,5]. Due to the high temperature of these clouds, the formation of the first objects via gravitational collapse would not be possible in the absence of the molecular cooling. The cooling process occurs via ro-vibrational transitions of such primordial molecules as H_2, H_2^+, HeH^+, LiH, LiH^+, and H_3^+, and their respective isotopologues. With the largest permanent dipole moment among the primordial molecules, LiH is particularly important for the cooling process [4]. The chemistry of Li compounds in the primordial universe was described by Bougleux and Galli [6] and by Lepp et al. [7].

The spectroscopy of LiH was studied in detail by Dulick et al. [8]. The PECs of the $X^1\Sigma^+$ and $A^1\Sigma^+$ states of LiH were empirically determined by Coxon and Dickson [9] from a large set of spectral lines of the four isotopologues of this system. An empirical PEC for the LiH ground state was also derived by Chan et al. [10]. One of the goals of those two works was to investigate the post-Born–Oppenheimer effects in the LiH rovibrational transitions.

In a recent work, where a review of the previous studies of the LiH rovibrational spectrum can be found, Holka et al. [3] obtained a highly accurate PEC from MR-CISD (multi-reference configuration-interaction single–double) electronic calculations that included size-extensivity, relativistic, and diagonal adiabatic corrections. The non-adiabatic effects in the energies of the vibrational levels were included using the *ab initio* formalism of Bunker and Moss [11]. The calculated rovibrational transitions obtained at this level of calculations differed from the experimental values by 1–2 cm^{-1}. In another work, Tung et al. [2] employed all-electron explicitly

* Corresponding author.
E-mail addresses: leogabriel@deii.cefetmg.br (L.G. Diniz), kirnosov@email.arizona.edu (N. Kirnosov), alexander.alijah@univ-reims.fr (A. Alijah), rachid@fisica.ufmg.br (J.R. Mohallem), ludwik@u.arizona.edu (L. Adamowicz).

http://dx.doi.org/10.1016/j.jms.2016.03.001
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correlated Gaussian functions with shifted centers to evaluate the non-relativistic PEC for LiH. Non-adiabatic corrections in that work were included in an approximate way by using the atomic masses instead of the nuclear masses in the calculation of the vibrational frequencies (though this was not explicitly mentioned in their paper). Even with this highly accurate PEC, the differences between the calculated and the experimental vibrational transition energies were of the order of 1–2 cm^{-1}, which undoubtedly resulted from an inadequate account of the non-adiabatic effects. Recently, a new approach for accounting for the non-adiabatic effects was proposed by Diniz et al. [12]. In the approach the reduced vibrational mass of the system was made dependent on the internuclear distance, R, of the molecule. The application of the approach in the calculations of the vibrational transition energies of LiH allowed for increasing the accuracy of the results to the level of about 0.2 cm^{-1}.

With the increased accuracy of the vibrational transitions achieved with the R-dependent effective reduced-mass approach a question appeared whether the use of the approach would also lead to higher accuracy in predicting other spectroscopic quantities such as the transition dipole moments and the transition probabilities, also called Einstein coefficients. Elucidation of this point is one of the goals of the present work. The Einstein coefficients have fundamental importance in the study of the cooling of primordial clouds. The accuracy of the evaluation of this property depends not only on the accuracy of the PEC but also on the accuracy of the dipole moment curve (DMC). As far as we know, the only available DMC of LiH was obtained in 1981 in the CI calculations by Partridge and Langhoff [13]. More recently Tunega and Noga [14] studied electrical properties of LiH using the MBPT-R12 approach but the dipole moment in that study was only calculated at the equilibrium distance. In the present work all-electron explicitly correlated Gaussians with shifted centers are employed in the calculation of the LiH DMC. With that the resulting DMC is likely the most accurate to date for this molecule. Details of the calculations are presented in Section 2.

As far as we know, there are only two recent reports on the complete ro-vibrational linelist of LiH in the literature. In the first one, published in 2011, Coppola et al. [4] used their own PEC and DMC obtained with the internally contracted multi-reference averaged-pair functional (IC-ACPF) calculations. The calculations were performed with the basis set consisting of the aug-cc-pVnZ (n = T,Q,5) Dunning basis set for the hydrogen atom [15] and the aug-cc-pwCnZ(-DK) (n = T,Q,5) basis set of Prascher et al. for the lithium atom [16]. In the calculations of the second linelist published in 2013 by Shi et al. [5], the empirical Coxon and Dickinson PEC [9] and the theoretical Partridge and Langhoff DMC [13] were used. Although the latter authors used a very accurate PEC generated from fitting the experimental spectral lines, the DMC they used was less accurate. Due to these inaccuracies it is difficult to determine which of the two linelists is more accurate. However, as the present calculations are performed with very accurate PEC and DMC, the relative quality of the two previous linelists can be evaluated. The present calculations are performed for the rovibrational transitions corresponding to $\Delta J = -1$ and $\Delta v \leqslant 5$.

Another interesting point which the present calculations enable to elucidate is how the non-adiabatic effects affect the values of the Einstein coefficients. Both Shi et al. [5] and Coppola et al. [4] included these effects in the calculations of the rovibrational frequencies by in an effective way by using the atomic reduced mass instead of the nuclear reduced mass. In the present calculations we also use these masses and we compare the results with the results obtained with the approach of Diniz et al. [12] that employs an R-dependent effective reduced mass. That approach, as mentioned, produced the most accurate rovibrational transition energies as compared to the experimental values.

In Section 2 the method used in the present calculations of the DMC is described and some computational details are discussed. In Section 3 the approach used to solve the rovibrational problem and to calculate the transition probabilities is presented. Section 4 is dedicated to the results and discussion. Finally, in Section 5 the main conclusions of this work are presented.

2. Computation of the dipole moment

In the present calculations the spatial component of the electronic wave function of the LiH molecule, $\Phi_M(\mathbf{r})$, where $\mathbf{r}$, is a 12-dimensional vector of the spatial coordinates of the four electrons, is expanded in terms of a set of M explicitly correlated all-electron Gaussians (ECGs), $\{g_k\}_{k=1,\ldots,M}$, with shifted centers as:

$$\Phi_m(\mathbf{r}) = \sum_{k=1}^{M} c_k \mathcal{P} g_k(\mathbf{r}), \tag{1}$$

where $\mathcal{P}$ is the operator that imposes the right permutational symmetry of the wave function, c_k are the linear expansion coefficients, and g_k are ECGs. They are the following 12-dimensional functions:

$$g_k(\mathbf{r}) = \exp\left[-(\mathbf{r}-\mathbf{s}_k)'\mathbf{A}_k(\mathbf{r}-\mathbf{s}_k)\right], \tag{2}$$

where $\mathbf{s}_k$ is a 12-dimensional vector of the Cartesian coordinates of the Gaussian shifts, prime denotes vector transposition, and $\mathbf{A}_k$ is a 12×12 dimensional symmetric matrix of the Gaussian exponential parameters defined as:

$$\mathbf{A}_k = A_k \otimes \mathbf{I}_3, \tag{3}$$

with $\mathbf{I}_3$ being a 3×3 identity matrix, $\otimes$ denoting the Kronecker product, and A_k being a 4×4 symmetric matrix. As in the present calculations, the LiH molecule is placed along the z axis, the Gaussian shifts are also restricted to this axis (the x and y components of the $\mathbf{S}_k$ vectors are set to zero).

For Gaussian (2) to be square integrable, as a basis function used for expanding the wave function of a bound electronic state should be, A_k has to be positive definite. This automatically happens if A_k is represented in the Cholesky-factored form as $L_k L_k'$ with L_k being a lower triangular matrix whose elements are real numbers ranging from $-\infty$ to $+\infty$.

The so-called spin-free formalism [17,18] is used to construct the permutation symmetry operator, $\mathcal{P}$. In this formalism, an appropriate symmetry projector is applied to the spatial parts of the wave function to impose the desired symmetry properties. The symmetry projector is constructed using the standard procedure involving Young operators as described, for example, in Ref. [19]. In the case of the singlet ground electronic state of the LiH molecule the permutation operator, $\mathcal{P}$, used in the present calculations has the following form: $\mathcal{P} = (1 - P_{13})(1 - \hat{P}_{24})(1 + \hat{P}_{12})(1 + \hat{P}_{34})$, where $\hat{P}_{ij}$ denotes the permutation of the spatial coordinates of the ith and jth electrons. As the electronic Hamiltonian is invariant upon any permutation of the electron labels, in the calculation of a Hamiltonian or overlap matrix element the permutation operators can be moved to either the basis function in the *bra* or the *ket*. The combined symmetry operator, $\mathcal{P}^\dagger\mathcal{P}$ consists of $4! = 24$ terms. Thus, each matrix elements is a sum of 24 contributions corresponding to the different components of the permutation operator.

The linear coefficients, c_k, in the expansion of the wave function in terms of the basis functions and the nonlinear parameters of the basis functions, *i.e.*, the exponent matrices, L_k and the shifts, $\mathbf{s}_k$, are obtained using the variational method by the minimization of the functional representing the total energy of the system:

$$E(\{L_k\},\{\mathbf{s}_k\},\{c_k\}) = \min_{\{L_k\},\ \{\mathbf{s}_k\},\{c_k\}} \frac{c'H(\{L_k,\},\{\mathbf{s}_k\})c}{c'S(\{L_k\},\{\mathbf{s}_k\})c}, \tag{4}$$

where $H(\{L_k\},\{\mathbf{s}_k\})$and $S(\{L_k\},\{\mathbf{s}_k\})$ are the Hamiltonian and overlap matrices, respectively. Both matrices depend on the nonlinear parameters of the basis functions. c is a column vector of the c_k coefficients.

To achieve high accuracy in the calculation of each PEC point, the linear and non-linear parameters of the Gaussians have to be extensively optimized. As each Gaussian contains 14 non-linear parameters, there are $14M$ non-linear optimization parameters in the basis set of M ECGs. To expedite this optimization the analytic gradient of the energy determined with respect to the Gaussian nonlinear parameters is used [2]. The use of the analytical gradient is key in achieving high accuracy and in lowering the computational cost of the PEC calculation. More information about the approach used in growing the basis set to the size of 2000 ECGs for the PEC point corresponding to the equilibrium internuclear distance and about optimizing the basis set for each other PEC point can be found in Ref. [2].

The ECG wave functions generated before in the LiH PEC calculations [2] are used in this work to calculate a dipole moment curve (DMC). The dipole moment vector, $\boldsymbol{\mu}$, for each PEC point is determined as the expectation value of the dipole operator as:

$$\boldsymbol{\mu} = \sum_{k,l}^{M} c_k c_l \langle \mathcal{P} g_k(\mathbf{r}) | \left(-\sum_{i=1}^{4} \mathbf{r}_i \right) | \mathcal{P} g_l(\mathbf{r}) \rangle + \sum_{\alpha=1}^{2} Q_\alpha \mathbf{R}_\alpha \tag{5}$$

$$= \sum_{k,l}^{M} c_k c_l \langle g_k(\mathbf{r}) | \left(-\sum_{i=1}^{4} \mathbf{r}_i \right) | \mathcal{P}^\dagger \mathcal{P} g_l(\mathbf{r}) \rangle + \sum_{\alpha=1}^{2} Q_\alpha \mathbf{R}_\alpha, \tag{6}$$

where $\mathbf{r}_i$ is the coordinate vector of electron i and Q_α and $\mathbf{R}_\alpha$ are the charges and the coordinates of the nuclei, respectively. In Eq. (6), we use the fact that the electronic dipole-moment operator, $-\sum_{i=1}^{4}\mathbf{r}_i$, is symmetric with respect to the permutation of the electrons. Also, the spatial wave function is assumed to be normalized. The dipole moment matrix element involving two ECGs, $g_k(\mathbf{r})$ and $g_l(\mathbf{r})$, is given as:

$$\langle g_k(\mathbf{r}) | \mathbf{r}_i | g_l(\mathbf{r}) \rangle = S_{kl} \left\{ [\mathbf{A}_k + \mathbf{A}_L]^{-1} [\mathbf{A}_k \mathbf{s}_k + \mathbf{A}_l \mathbf{s}_l] \right\}_i, \tag{7}$$

where $\mathbf{v}_i$ denotes the ith component of vector $\mathbf{v}$ and $S_{kl} = \langle g_k | g_l \rangle$ is the overlap integral.

3. Rovibrational eigenvalues and transition probabilities

The radial (vibrational) Schrödinger equation within the adiabatic approximation for a diatomic molecule is:

$$\left[-\frac{1}{2\mu_v} \frac{d^2}{dR^2} + E_e(R) + \frac{J(J+1)}{2\mu_r R^2} - E_{vJ} \right] \chi_{vJ}(R) = 0, \tag{8}$$

where R is the internuclear distance, μ_v is the vibrational reduced mass, μ_r is the rotational reduced mass, $E_e(R)$ is the PEC obtained in the electronic structure calculations, and χ_{vJ} is the vibrational wave function corresponding to the vibrational quantum numbers v and the rotational quantum number J. In the adiabatic approximation μ_v and μ_r are equal to the nuclear reduced mass. As mentioned, the LiH PEC used in the present work is take from the work of Tung et al. [2]. It includes the Born–Oppenheimer energy curve calculated on a grid of points with 2000 fully optimized explicitly correlated Gaussians with shifted centers, the diagonal adiabatic correction (DBOC) curve, and the leading relativistic correction curve taken from the work of Holka et al. [3]. In order to improve the numerical stability of the calculations of the higher vibrational states the PEC is extended to some distances smaller than the smallest distance of $1.8\,a_0$ reported by Tung et al. [2]. The extension is done through extrapolation involving fitting the PEC in the interval between $1.8\,a_0$ and $2.5\,a_0$ with function $f(R) = a + b\exp(cR)$ and using the function to determine the PEC points for the distances smaller than $1.8\,a_0$. The fitting results in the following values of the adjustable parameters involved in the fitting function: $a = (-8.0852 \pm 0.0009)E_h$, $b = (7.8 \pm 0.4)E_h$, and $c = (-2.26 \pm 0.03)a_0^{-1}$. The PEC is extrapolated down to $R = 1.2\,a_0$.

The Einstein probabilities are determined using the DMC according to the following expression:

$$A_{v'J' \to v''J''} = \left(\frac{16\pi^3}{3\epsilon_0 h} \right) \frac{S(J', J'')}{2J'+1} \nu^3 \, |\langle \chi_{v'J'} | D(R) | \chi_{v''J''} \rangle|^2, \tag{9}$$

where

$$S(J', J'') = \begin{cases} J'' & \text{if } J'' - J' = -1 \quad (P-\text{branch}) \\ J''+1 & \text{if } J'' - J' = 1 \quad (R-\text{branch}) \end{cases} \tag{10}$$

is the Hönl-London rotational intensity factor, ν is the transition frequency, and $D(R)$ is the DMC. With ν given in cm^{-1} and $D(R)$ in Debye, the constant in the parenthesis in Eq. (9) is equal to $3.1361891 \times 10^{-7} \text{cm}^3\ \text{D}^{-2}\ \text{s}^{-1}$.

The radial vibrational Eq. (8) is solved with the renormalized Numerov code [20] adapted to allow for a R-dependent reduced mass [21]. The numerical integration is performed in the range $1.2\,a_0 \leqslant R \leqslant 30.0\,a_0$. The convergence criterion for the eigenvalues is $10^{-14}\,E_h$. The non-adiabatic effects in the calculation of rovibrational levels away from avoided crossings can be accounted for in an approximate way by replacing the nuclear vibrational reduced mass, μ_v in Eq. (8), with an effective R-dependent reduced mass, $\mu_v(R)$. Recently, Diniz et al. [12] proposed a model for determining the R-dependent vibrational reduced mass that significantly improves the accuracy of the calculated vibrational levels of LiH in comparison with the results obtained using the conventional approach where the vibrational mass is taken as the atomic reduced mass and rotational mass is taken as the nuclear reduced mass [22]. Due to the use of an R-dependent vibrational reduced mass, the rovibrational wavefunctions for a given J value are not exactly orthogonal. However, as we have verified numerically, the overlap between different wavefunctions is almost negligible, of the order of 10^{-5}. There is no noticeable effect of this non-orthogonality on the computed intensities. The conventional model and the Diniz et al. model are used in the present calculations.

4. Results and discussion

The Einstein A coefficients, the transition frequencies, and the transition dipole moments for rovibrational transitions $\Delta J = -1$ and $\Delta v \leqslant 5$ are presented in Table 1. The present values (of the Einstein coefficients) are compared in the table with the values from the two recently published linelists by Coppola et al. [4] (A_{Cop}) and by Shi et al. [5] (A_{Shi}). As mentioned, the Coppola et al. linelist calculations involved their own potential energy and dipole-moment curves, and in the linelist calculations by Shi et al. [5] the empirical potential energy curve derived from the experimental data by Coxon and Dickinson [9] and the dipole moment curve of Partridge and Langhoff [13] were used. While the two linelists were obtained with the use of the atomic reduced mass in the rovibrational calculation, our results are obtained with three different reduced masses, i.e., the nuclear mass, the atomic mass, and the effective R-dependent vibrational mass of Diniz et al. [12] (the reduced masses are provided in the supplemental material). The subscripts nu, at and eff in the table indicate the nuclear, atomic, and effective masses, respectively. Since the rotational non-adiabatic effects are much smaller than the vibrational ones, in calculating states with low J values the rotational mass can be left equal to the reduced nuclear mass.

Table 1
Transition frequencies (v), Dipole moments (D) and Einstein coefficients (A) for the $\Delta J = -1$ and $\Delta v \leq 5$ rovibrational transitions of ^{7}LiH. The experimental transition frequencies (v_{exp}) were obtained from Coxon and Dickinson's empirical quantum mechanical molecular constants [9] (see Eq. (17) in this reference). The subscripts *nu*, *at* and *eff* indicate respectively the use of nuclear, atomic and effective (i.e. $\mu_v(R)$) reduced mass in the rovibrational calculations. $\Delta v_{mass} = v_{mass} - v_{exp}$. Einstein coefficients are compared with those obtained by Coppola et al. [4] (A_{Cop}) and by Shi et al. [5] (A_{Shi}).

Upper		Lower		Transition frequencies (cm^{-1})				Dipole moments ($e a_0$)			Einstein coefficients (s^{-1})				
v'	J'	v''	J''	v_{exp}	Δv_{nu}	Δv_{at}	Δv_{eff}	D_{nu}	D_{at}	D_{eff}	A_{nu}	A_{at}	A_{eff}	A_{Cop}	A_{Shi}
1	0	0	1	1344.900	0.452	0.127	−0.039	−0.24775	−0.24772	−0.24778	46.875	46.828	46.834	46.619	45.392
2	0	0	1	2659.752	0.873	0.238	−0.077	−0.01875	−0.01875	−0.01877	2.077	2.074	2.078	2.101	1.989
3	0	0	1	3930.661	1.276	0.352	−0.091	−0.00070	−0.00070	−0.00071	0.009	0.009	0.010	0.010	0.002
4	0	0	1	5158.468	1.654	0.463	−0.107	0.00023	0.00023	0.00023	0.002	0.002	0.002	0.002	0.004
5	0	0	1	6343.919	2.009	0.572	−0.115	0.00016	0.00016	0.00016	0.002	0.002	0.002	0.002	0.001
2	0	1	1	1300.468	0.408	0.105	−0.052	−0.35250	−0.35245	−0.35252	85.787	85.704	85.709	85.262	82.763
3	0	1	1	2571.378	0.811	0.219	−0.066	−0.03490	−0.03489	−0.03491	6.500	6.491	6.498	6.568	6.435
4	0	1	1	3799.184	1.189	0.331	−0.081	−0.00178	−0.00178	−0.00180	0.055	0.054	0.056	0.059	0.052
5	0	1	1	4984.636	1.544	0.439	−0.090	0.00042	0.00042	0.00041	0.007	0.007	0.007	0.005	0.007
6	0	1	1	6128.380	1.862	0.530	−0.097	0.00034	0.00034	0.00035	0.008	0.008	0.009	0.008	0.005
3	0	2	1	1256.943	0.390	0.108	−0.027	−0.43272	−0.43266	−0.43276	116.724	116.614	116.631	115.930	112.536
4	0	2	1	2484.749	0.768	0.219	−0.042	−0.05294	−0.05292	−0.05294	13.495	13.478	13.483	13.629	13.395
5	0	2	1	3670.201	1.123	0.327	−0.051	−0.00350	−0.00349	−0.00350	0.190	0.189	0.190	0.195	0.179
6	0	2	1	4813.945	1.441	0.419	−0.058	0.00051	0.00051	0.00050	0.009	0.009	0.009	0.007	0.010
7	0	2	1	5916.507	1.742	0.514	−0.053	0.00056	0.00056	0.00056	0.021	0.021	0.020	0.020	0.022
4	0	3	1	1214.250	0.366	0.105	−0.028	−0.49879	−0.49872	−0.49882	139.813	139.686	139.695	138.710	134.464
5	0	3	1	2399.702	0.721	0.213	−0.037	−0.07321	−0.07319	−0.07321	23.249	23.219	23.229	23.487	23.125
6	0	3	1	3543.446	1.039	0.305	−0.044	−0.00588	−0.00587	−0.00588	0.482	0.481	0.483	0.487	0.445
7	0	3	1	4646.008	1.340	0.400	−0.039	0.00049	0.00049	0.00048	0.007	0.007	0.007	0.005	0.008
8	0	3	1	5707.769	1.632	0.506	−0.018	0.00083	0.00083	0.00082	0.040	0.040	0.039	0.036	0.041
5	0	4	1	1172.299	0.343	0.102	−0.022	−0.55395	−0.55388	−0.55399	155.179	155.045	155.060	153.740	148.733
6	0	4	1	2316.043	0.661	0.194	−0.029	−0.09595	−0.09591	−0.09594	35.899	35.852	35.860	36.322	35.868
7	0	4	1	3418.605	0.962	0.289	−0.024	−0.00900	−0.00900	−0.00902	1.016	1.015	1.020	1.023	0.942
8	0	4	1	4480.367	1.254	0.394	−0.003	0.00020	0.00020	0.00019	0.001	0.001	0.001	0.001	0.003
9	0	4	1	5501.533	1.517	0.492	0.023	0.00102	0.00102	0.00102	0.054	0.054	0.054	0.051	0.058
6	0	5	1	1130.990	0.306	0.086	−0.020	−0.59923	−0.59916	−0.59930	163.047	162.915	162.944	161.250	155.587
7	0	5	1	2233.551	0.607	0.181	−0.015	−0.12158	−0.12154	−0.12155	51.699	51.631	51.631	52.297	51.763
8	0	5	1	3295.313	0.899	0.286	0.006	−0.01302	−0.01301	−0.01303	1.903	1.899	1.905	1.909	1.787
9	0	5	1	4316.480	1.163	0.384	0.032	−0.00024	−0.00024	−0.00025	0.001	0.001	0.002	0.002	0.000
10	0	5	1	5297.039	1.407	0.483	0.069	0.00119	0.00118	0.00118	0.066	0.065	0.065	0.061	0.072
7	0	6	1	1090.200	0.290	0.089	−0.007	−0.63450	−0.63444	−0.63460	163.731	163.609	163.646	161.570	155.398
8	0	6	1	2151.962	0.582	0.195	0.014	−0.15024	−0.15018	−0.15020	70.606	70.513	70.512	71.549	70.941
9	0	6	1	3173.128	0.845	0.293	0.039	−0.01808	−0.01807	−0.01808	3.278	3.272	3.276	3.277	3.106
10	0	6	1	4153.688	1.089	0.392	0.077	−0.00107	−0.00107	−0.00109	0.026	0.026	0.027	0.027	0.016
11	0	6	1	5093.349	1.291	0.470	0.105	0.00125	0.00125	0.00124	0.065	0.065	0.064	0.061	0.077
8	0	7	1	1049.789	0.281	0.100	0.009	−0.65903	−0.65898	−0.65915	157.711	157.608	157.648	155.140	148.622
9	0	7	1	2070.956	0.545	0.198	0.034	−0.18249	−0.18241	−0.18244	92.839	92.717	92.722	94.170	93.556
10	0	7	1	3051.515	0.789	0.297	0.072	−0.02435	−0.02433	−0.02434	5.287	5.278	5.281	5.287	5.044
11	0	7	1	3991.176	0.990	0.376	0.100	−0.00227	−0.00227	−0.00229	0.103	0.103	0.104	0.104	0.081
12	0	7	1	4889.289	1.162	0.448	0.133	0.00119	0.00119	0.00117	0.052	0.052	0.050	0.046	0.061
9	0	8	1	1009.581	0.253	0.092	0.013	−0.67128	−0.67126	−0.67144	145.533	145.455	145.499	142.590	135.858
10	0	8	1	1990.141	0.497	0.191	0.051	−0.21869	−0.21860	−0.21863	118.316	118.160	118.168	120.150	119.639
11	0	8	1	2929.801	0.699	0.270	0.079	−0.03210	−0.03208	−0.03210	8.135	8.120	8.125	8.162	7.838
12	0	8	1	3827.914	0.871	0.343	0.112	−0.00395	−0.00395	−0.00396	0.275	0.274	0.276	0.272	0.229
13	0	8	1	4683.373	1.058	0.456	0.199	0.00091	0.00091	0.00089	0.026	0.027	0.025	0.023	0.033
10	0	9	1	969.359	0.233	0.093	0.026	−0.66916	−0.66918	−0.66938	128.007	127.958	128.007	124.740	117.980
11	0	9	1	1909.020	0.435	0.172	0.054	−0.25937	−0.25925	−0.25927	146.879	146.688	146.687	149.260	148.928
12	0	9	1	2807.133	0.607	0.245	0.087	−0.04179	−0.04176	−0.04178	12.123	12.101	12.109	12.223	11.828
13	0	9	1	3662.592	0.794	0.358	0.174	−0.00626	−0.00626	−0.00627	0.605	0.603	0.605	0.582	0.510
14	0	9	1	4473.706	0.979	0.498	0.303	0.00029	0.00029	0.00028	0.002	0.002	0.002	0.002	0.006
11	0	10	1	928.847	0.192	0.073	0.016	−0.64984	−0.64991	−0.65014	106.198	106.181	106.236	102.740	96.167
12	0	10	1	1826.959	0.364	0.146	0.050	−0.30507	−0.30492	−0.30493	178.088	177.859	177.842	180.880	180.783
13	0	10	1	2682.418	0.551	0.259	0.137	−0.05408	−0.05404	−0.05406	17.715	17.681	17.691	17.971	17.531
14	0	10	1	3493.532	0.735	0.399	0.265	−0.00913	−0.00911	−0.00912	1.114	1.111	1.114	1.100	0.985
15	0	10	1	4257.860	0.925	0.579	0.450	−0.00074	−0.00074	−0.00074	0.013	0.013	0.013	0.011	0.006
12	0	11	1	887.689	0.163	0.067	0.022	−0.60958	−0.60973	−0.61000	81.560	81.574	81.635	78.096	71.991
13	0	11	1	1743.147	0.350	0.181	0.109	−0.35576	−0.35559	−0.35558	210.364	210.102	210.069	213.640	213.751
14	0	11	1	2554.262	0.534	0.321	0.238	−0.07026	−0.07020	−0.07021	25.816	25.765	25.773	26.228	25.826
15	0	11	1	3318.589	0.724	0.500	0.422	−0.01300	−0.01298	−0.01300	1.938	1.933	1.937	1.919	1.741

(*continued on next page*)

Table 1 (*continued*)

Upper		Lower		Transition frequencies (cm^{-1})				Dipole moments (ea_0)			Einstein coefficients (s^{-1})				
v'	J'	v''	J''	v_{exp}	Δv_{nu}	Δv_{at}	Δv_{eff}	D_{nu}	D_{at}	D_{eff}	A_{nu}	A_{at}	A_{eff}	A_{Cop}	A_{Shi}
16	0	11	1	4032.719	0.909	0.717	0.664	−0.00210	−0.00210	−0.00210	0.091	0.090	0.091	0.093	0.073
13	0	12	1	845.431	0.178	0.108	0.076	−0.54401	−0.54427	−0.54460	56.122	56.160	56.223	52.841	47.533
14	0	12	1	1656.545	0.362	0.248	0.204	−0.41125	−0.41105	−0.41103	241.268	240.986	240.941	244.810	244.937
15	0	12	1	2420.873	0.552	0.428	0.389	−0.09177	−0.09168	−0.09168	37.497	37.419	37.419	38.334	38.168
16	0	12	1	3135.002	0.737	0.644	0.631	−0.01841	−0.01839	−0.01840	3.278	3.269	3.275	3.224	2.958
17	0	12	1	3794.244	0.793	0.783	0.817	−0.00415	−0.00414	−0.00415	0.295	0.294	0.295	0.296	0.251
14	0	13	1	801.494	0.175	0.135	0.117	−0.44711	−0.44751	−0.44793	32.302	32.354	32.412	29.552	25.434
15	0	13	1	1565.822	0.365	0.314	0.302	−0.47030	−0.47008	−0.47005	266.492	266.216	266.174	269.630	269.228
16	0	13	1	2279.951	0.551	0.531	0.544	−0.12126	−0.12113	−0.12111	54.695	54.574	54.554	56.328	56.723
17	0	13	1	2939.193	0.606	0.669	0.730	−0.02628	−0.02624	−0.02625	5.502	5.486	5.492	5.471	5.129
18	0	13	1	3537.150	0.359	0.569	0.708	−0.00716	−0.00714	−0.00715	0.711	0.708	0.711	0.674	0.576
15	0	14	1	755.133	0.181	0.174	0.174	−0.31183	−0.31242	−0.31293	13.141	13.190	13.233	11.293	8.754
16	0	14	1	1469.262	0.367	0.391	0.415	−0.52906	−0.52884	−0.52881	278.632	278.412	278.400	280.460	278.765
17	0	14	1	2128.504	0.422	0.529	0.601	−0.16271	−0.16251	−0.16245	80.112	79.929	79.878	82.731	84.026
18	0	14	1	2726.461	0.175	0.429	0.579	−0.03915	−0.03908	−0.03908	9.744	9.711	9.711	9.864	9.560
19	0	14	1	3254.425	−0.266	0.216	0.479	−0.01135	−0.01132	−0.01134	1.392	1.386	1.390	1.349	1.170
16	0	15	1	705.384	0.177	0.211	0.230	−0.13056	−0.13139	−0.13202	1.878	1.902	1.920	1.208	0.496
17	0	15	1	1364.626	0.233	0.349	0.417	−0.57905	−0.57888	−0.57889	267.362	267.270	267.318	267.030	263.330
18	0	15	1	1962.583	−0.014	0.250	0.394	−0.22004	−0.21975	−0.21964	114.780	114.529	114.443	118.720	120.897
19	0	15	1	2490.547	−0.455	0.037	0.294	−0.06257	−0.06243	−0.06239	18.955	18.882	18.863	19.491	19.572
20	0	15	1	2936.809	−0.609	0.216	0.631	−0.01930	−0.01925	−0.01925	2.958	2.945	2.946	2.920	2.698
17	0	16	1	650.982	0.048	0.133	0.176	0.10484	0.10371	0.10295	0.951	0.931	0.918	1.515	2.546
18	0	16	1	1248.939	−0.198	0.033	0.153	−0.60403	−0.60402	−0.60411	222.814	222.925	223.059	219.610	213.996
19	0	16	1	1776.903	−0.639	−0.180	0.054	−0.29510	−0.29472	−0.29458	153.058	152.790	152.698	157.620	159.673
20	0	16	1	2223.165	−0.794	−0.001	0.390	−0.10598	−0.10572	−0.10561	38.666	38.514	38.455	40.367	41.259
21	0	16	1					−0.03863	−0.03849	−0.03845	7.968	7.922	7.907	8.115	8.142
18	0	17	1	590.232	−0.251	−0.103	−0.031	0.39944	0.39798	0.39707	10.276	10.209	10.166	11.646	13.863
19	0	17	1	1118.196	−0.692	−0.316	−0.131	−0.57365	−0.57399	−0.57427	144.027	144.345	144.555	138.810	132.600
20	0	17	1	1564.458	−0.847	−0.137	0.206	−0.37736	−0.37700	−0.37687	170.724	170.635	170.628	173.090	173.115
21	0	17	1					−0.17821	−0.17780	−0.17762	69.798	69.604	69.521	72.961	74.498
22	0	17	1					−0.08159	−0.08132	−0.08120	20.772	20.688	20.651	21.971	23.208
19	0	18	1	520.844	−0.442	−0.213	−0.105	0.74675	0.74501	0.74398	24.647	24.565	24.513	25.972	28.392
20	0	18	1	967.106	−0.596	−0.034	0.231	−0.43716	−0.43820	−0.43880	54.113	54.465	54.659	50.320	45.809
21	0	18	1					−0.41831	−0.41840	−0.41848	125.219	125.564	125.746	123.320	120.245
22	0	18	1					−0.25196	−0.25181	−0.25174	74.679	74.828	74.893	75.481	77.729
23	0	18	1					−0.10423	−0.10472	−0.10488	15.633	15.847	15.927	16.133	−99.900
20	0	19	1	439.847	−0.154	0.179	0.331	1.11103	1.10928	1.10826	32.908	32.879	32.853	33.586	35.156
21	0	19	1					−0.13891	−0.14091	−0.14194	2.980	3.076	3.125	2.351	1.497
22	0	19	1					−0.29718	−0.29854	−0.29918	29.997	30.396	30.582	28.516	27.295
23	0	19	1					−0.15587	−0.15745	−0.15809	11.137	11.429	11.550	10.901	
21	0	20	1					1.40463	1.40356	1.40287	25.282	25.346	25.366	25.344	25.644
22	0	20	1					0.30013	0.29775	0.29657	5.560	5.502	5.472	5.931	6.798
23	0	20	1					0.03416	0.03267	0.03197	0.120	0.111	0.106	0.158	
22	0	21	1					1.42280	1.42410	1.42443	8.001	8.079	8.110	7.961	8.226
23	0	21	1					0.48079	0.48296	0.48364	2.866	2.926	2.949	2.903	
23	0	22	1					0.76759	0.77473	0.77739	0.212	0.221	0.224	0.214	

Comparing the present results obtained with the three reduced masses with the experimental values, one can see some interesting features. As expected, the use of different reduced masses mass has an noticeable effect of the order of 1 cm^{-1} on the transition energies. The experimental values were calculated by the empirical quantum mechanical molecular constants (B_v, D_v, H_v, L_v, and M_v) of Coxon and Dickinson [9] (see Eq. (17) of this reference). As these constants were only reported for $v = 0 - 20$, the transitions involving $v > 20$ are not considered here. As shown in [12] and in the table, the use of the effective mass considerably improves the agreement between the theoretical and experimental transition energies. For transitions involving high vibrational levels, the agreement with the experimental values worsens significantly for all three masses. As mentioned in [12], for $v > 13$ the average values of R lie already in the avoided crossing region or beyond, so that can be the source of the larger errors. Furthermore, for these higher states, there is a discrepancy of approximately 0.5 cm^{-1} between the empirically adjusted experimental values of Chan et al. [10] and the values reported by Coxon and Dickinson [9]. Given this, additional measurements for higher vibrational levels and a full non-adiabatic theoretical treatment (not involving an effective-mass model) would be important to shed light on the issue.

In contrast to what happens with the transition energies, the effect of the different masses has rather negligible effect on the transition dipole moments. As it can be seen in the table, the values

obtained with the nuclear reduced mass differ very little from the values obtained with the atomic reduced mass. The values of the Einstein coefficients are also rather insensitive to the mass model used. The difference between the values obtained with the atomic mass and with the effective mass are of the order of only 0.01 s^{-1}. Thus, we conclude that, while the use the effective mass in calculating the transition energies is important, the simpler model involving the use of the atomic reduced mass in the calculations of the transition dipole moments and the transition probabilities is quite adequate.

The present calculations allow for assessing the accuracy of the two previously published LiH linelists [4,5]. The different PECs and DMCs used in those two calculations do not allow for an easy evaluation of which calculation is more accurate. While Coppola et al. [4] did not compare their results with any previous calculations, Shi et al. [5] compared their Einstein coefficients with those calculated by Coppola et al. and by Partridge and Langhoff [13]. In Table 2 we compare all previously calculated linelists with ours. The comparison shows that there is an approximately 3% difference between the values of the Einstein coefficient obtained in the different calculations. As the results of Shi et al. are obtained using an empirical PEC, their transition energies agree best with the experiment. However, as the DMC used by them is not empirical, it is unclear whether their transition probabilities are as accurate as their transition energies. Some clue concerning this point is obtained by comparing the transition probabilities of Shi et al. with our values, as the DMC used in the present calculations is definitely the most accurate ever calculated (the DMC is provided as supplemental material). In Fig. 1 we compare our DMC with the DMCs used in the two previous calculations. While the main figure shows the differences between those two curves and our curve, the inset shows the whole DMC obtained in the present calculations. The difference curves indicate that the DMC used by Coppola et al. agrees better with ours. This explains the better agreement of our Einstein coefficients with their values, as can be seen in Table 2.

Still the question remains whether despite the poorer quality of the DMC used by Shi et al., their use of a more accurate empirical PEC leads to more accurate results concerning the transition probabilities. To elucidate this point, we have calculated the Einstein coefficients using our PEC and the DMC of Shi et al. As the results of this calculation reproduce the values of Shi et al. (see the fifth column in the Table 2), it becomes clear that the quality of the DMC is what determines the accuracy of the Einstein coefficients. Thus, while the linelist by Shi et al. is more accurate in terms of the transition energies, the linelist from Coppola et al. is more accurate in terms of the transition probabilities.

The dipole moment for the LiH was measured [23] for the three lowest vibrational states and the $J = 1$ rotational state. By averaging our DMC over the wave functions of these vibrational states, a direct comparison between the calculated values and the experimental results can be made. A similar comparison can be made for the dipole moments obtained using the DMC of Partridge and Langhoff [13]. The comparison is shown in Table 3. While Partridge and Langhoff dipoles are outside the range of the experimental error, our results lie within this range. In Table 4 the vibrational averages of the dipole moment for the five lowest vibrational states and the $J = 0$ rotational state are presented. For the lowest state, the result is very close to the dipole moment value obtained

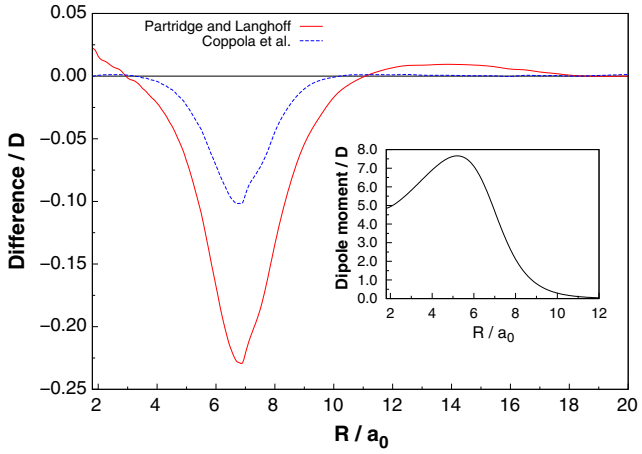


Fig. 1. Differences between the DMCs of Partridge and Langhoff [13] and of Coppola et al. [4] and the DMC obtained in this work (shown in the inset).

Table 3
Vibrationally averaged dipole moments (Debye) of ^{7}LiH in the ground $X^1\Sigma^+$ electronic state and in the $J = 1$ rotational state.

v	Partridge [13]	Nuclear	Atomic	Effective	Experiment [23]
0	5.889	5.884	5.884	5.884	5.882 ± 0.003
1	5.996	5.992	5.993	5.992	5.990 ± 0.003
2	6.104	6.101	6.101	6.101	6.098 ± 0.003

Table 4
Vibrationally averaged dipole moments (Debye) of ^{7}LiH in the ground $X^1\Sigma^+$ electronic state and in the $J = 0$ rotational state.

v	Partridge [13]	Nuclear	Atomic	Effective	Cafiero [24]
0	5.888	5.883	5.883	5.883	5.882
1	5.995	5.992	5.992	5.992	–
2	6.103	6.100	6.100	6.100	–
3	6.210	6.208	6.208	6.208	–
4	6.312	6.313	6.313	6.313	–
5	6.408	6.416	6.416	6.416	–

Table 2
Einstein coefficients, A (s^{-1}), for some transitions of ^{7}LiH in the ground $X^1\Sigma^+$ electronic state.

v'	J'	v''	J''	Nuclear	Atomic	Effective	Shi [5]	Coppola [4]	Partridge [13]
1	0	0	1	46.9	46.8	46.8	45.4	46.6	43.1
2	0	0	1	2.1	2.1	2.1	2.0	2.1	1.9
2	0	1	1	85.8	85.7	85.7	82.8	85.3	78.5
3	0	1	1	6.5	6.5	6.5	6.4	6.6	6.3
3	0	2	1	116.7	116.6	116.6	112.5	115.9	106.7
4	0	2	1	13.5	13.5	13.5	13.4	13.6	13.3
4	0	3	1	139.8	139.7	139.7	134.5	138.7	127.3
5	0	3	1	23.2	23.2	23.2	23.1	23.5	22.7
5	0	4	1	155.2	155.0	155.1	148.7	153.7	140.3
6	0	4	1	35.9	35.9	35.9	35.9	36.3	34.5
6	0	5	1	163.0	162.9	162.9	155.6	161.3	145.7
7	0	5	1	51.7	51.6	51.6	51.8	52.3	49.3

by Cafiero et al. [24] in the calculations where the Born–Oppenheimer approximation was not assumed.

5. Conclusions

Summarizing, a very accurate dipole moment and potential-energy curves together with an effective model to simulate the non-adiabatic effects are used in the calculations of the rovibrational spectrum of the ^{7}LiH molecule. The results show that, while the non-adiabatic effects are important for the transition energies, their effect on the transition dipole moments and the transition probabilities is negligible. The computed vibrationally averaged dipole moments are found to be in very good agreement with the experiment. The present very accurate results are used to assess the quality of the two complete rovibrational linelists available for ^{7}LiH in the literature. The results suggest that, despite the higher accuracy of the transition energies in the linelist of Shi et al. [5] as compared to those of Coppola et al. [4], the Einstein coefficients in the latter linelist are more accurate.

Acknowledgments

We acknowledge CNPq and Capes for a financial support (PEV Grant 401846/2013-0), the ROMEO HPC Center at the University of Reims Champagne-Ardenne and COST action CM 1405 Molecules in Motion (MOLIM). This work has been supported in part by the National Science Foundation with Grant #1444127. We thank J. Tennyson for providing their DMC (private communication).

Appendix A. Supplementary material

Supplemental material contains the dipole moment curve data used in the present calculations. Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jms.2016.03.001.

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